Intro to Neural Networks & Mechanistic Interpretability

Maths, Physics, and AI Alignment

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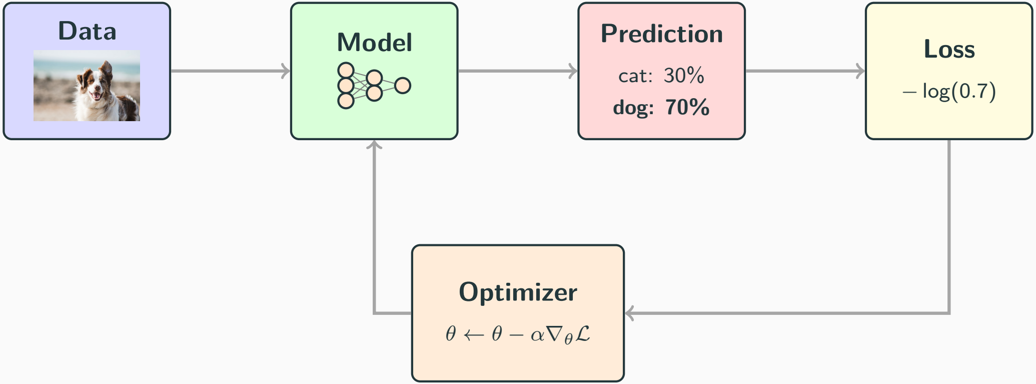
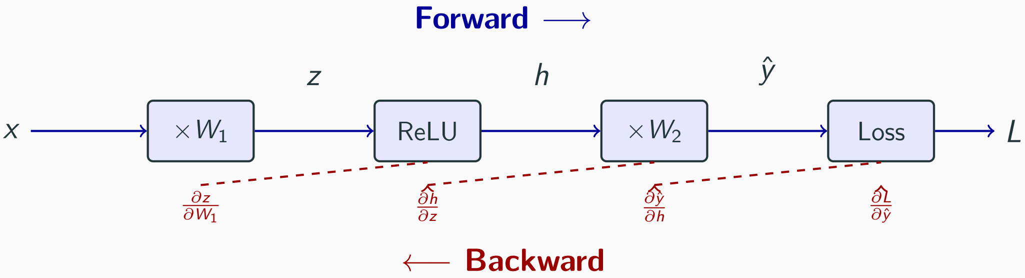
Overview

Machine Learning

The Training Loop



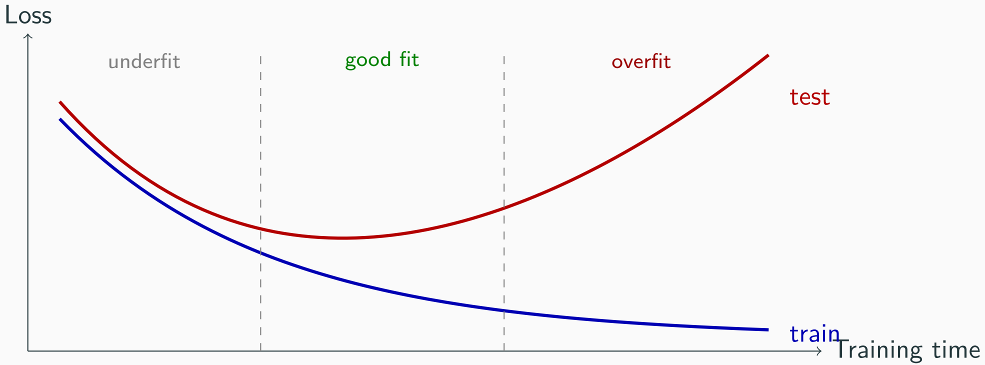
Forward Pass / Backward Pass



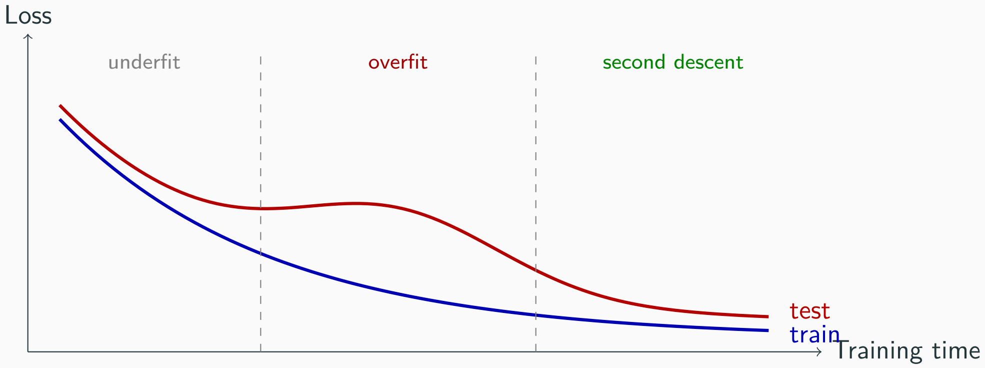
Chain rule:

$$\frac{\partial L}{\partial W_1} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial h} \cdot \frac{\partial h}{\partial z} \cdot \frac{\partial z}{\partial W_1}$$

Train Loss vs. Test Loss

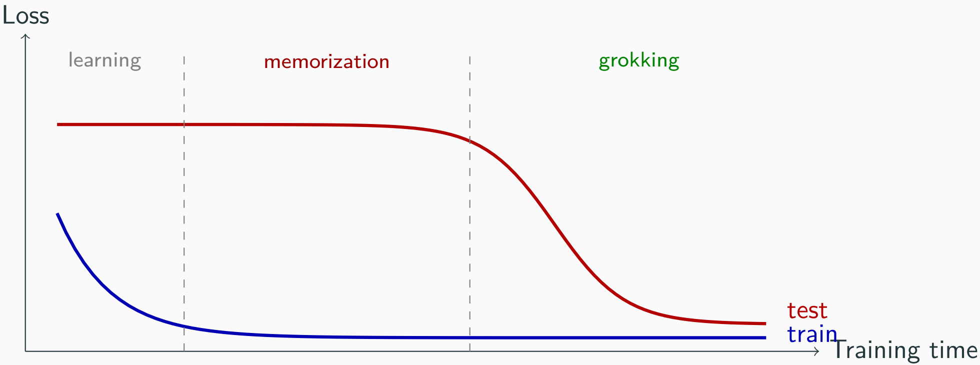


Double Descent



"Deep Double Descent: Where Bigger Models and More Data Can Hurt" [Nakkiran et al., 2019]

Grokking



"Grokking: Generalization Beyond Overfitting on Small Algorithmic Datasets" [Power et al., 2022]

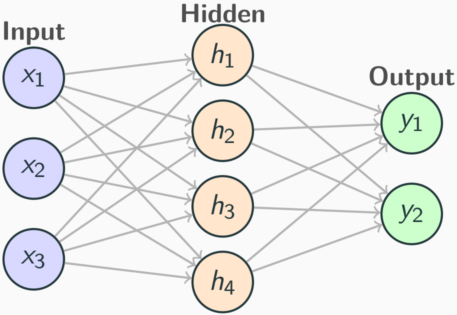
Definitions

Parameter / Weight the numbers that define the model, updated during training (θ, W, b)

Activation intermediate values computed during a forward pass

Hyperparameter choices made *before* training: learning rate, architecture, batch size

Multi-Layer Perceptron

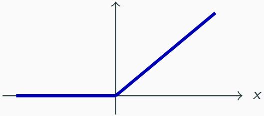


$\sigma(\cdot)$ between layers

$$h = \sigma(W_1 x + b_1)$$

$$y = W_2 h + b_2$$

ReLU



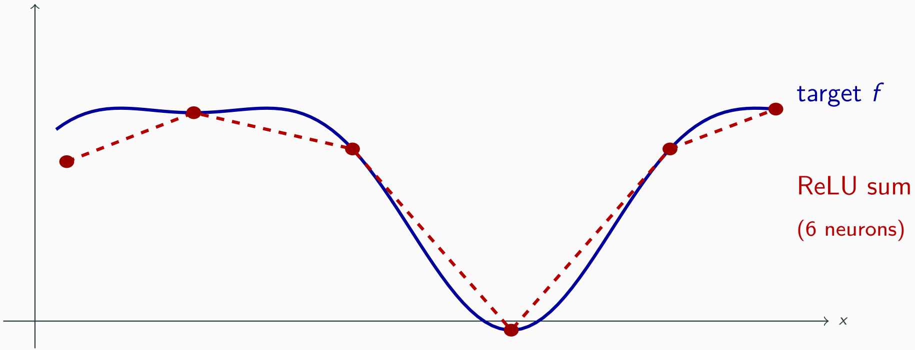
$\max(0, x)$

Universal Approximation Theorem

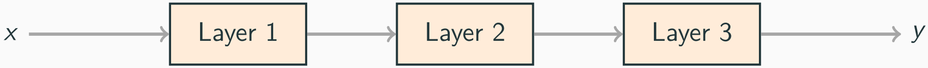
A neural network with a single hidden layer and a nonlinear activation function can approximate any continuous function on a compact set to arbitrary precision, given enough neurons.

"Approximation by Superpositions of a Sigmoidal Function" [Cybenko, 1989]

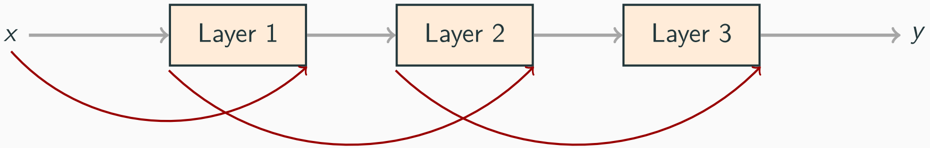
Universal Approximation: Intuition

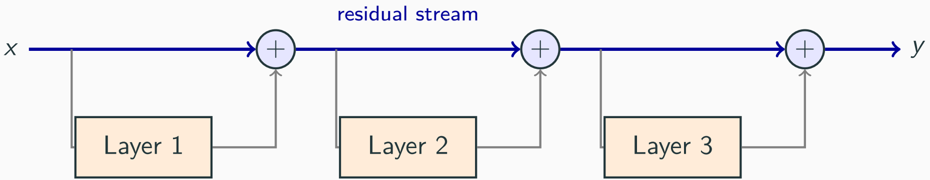


Layers

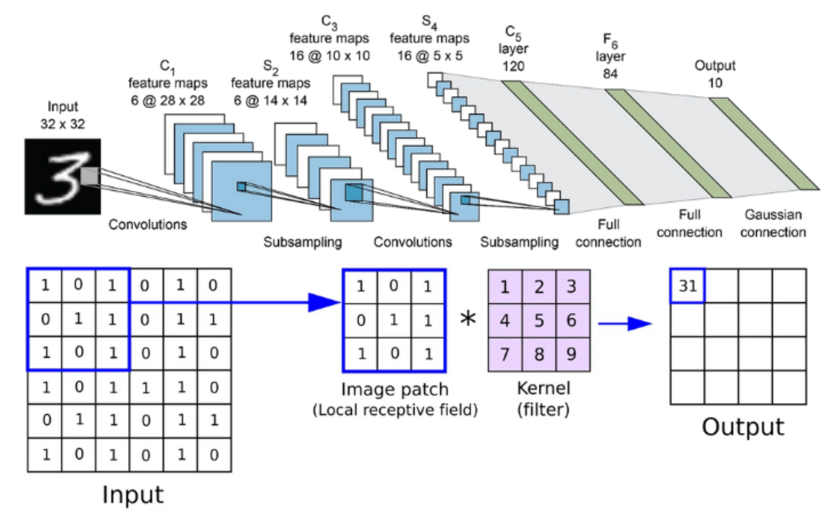


Residual Connections





Convolutional Neural Networks



"Gradient-Based Learning Applied to Document Recognition" [LeCun et al., 1998]

References i

Cybenko, George [1989]. “Approximation by Superpositions of a Sigmoidal Function”. In: *Mathematics of Control, Signals and Systems* 2.4.

LeCun, Yann, Léon Bottou, Yoshua Bengio, and Patrick Haffner [1998]. “Gradient-Based Learning Applied to Document Recognition”. In: *Proceedings of the IEEE* 86.11.

Nakkiran, Preetum, Gal Kaplun, Yamini Bansal, Tristan Yang, Boaz Barak, and Ilya Sutskever [2019]. “Deep Double Descent: Where Bigger Models and More Data Can Hurt”. In: *arXiv preprint arXiv:1912.02292*.

Power, Alethea, Yuri Burda, Harri Edwards, Igor Babuschkin, and Vedant Misra [2022]. “Grokking: Generalization Beyond Overfitting on Small Algorithmic Datasets”. In: *arXiv preprint arXiv:2201.02177*.